BLUEPRINT 2026

ScrapLink

Digitizing India’s Informal Waste Economy

Domain: Environment

1. Problem Statement

India generates over 62 million tonnes of waste annually. A significant portion of paper, plastic, metal, glass is recyclable. Yet the majority reaches landfills.

The reason is not a lack of infrastructure. India has 1.5 million informal scrap dealers **kabadiwalas** who already do last mile waste collection. They are the invisible backbone of India’s recycling economy.

The problem is coordination failure:

1) Households don’t know when their local kabadiwala will visit their area
2) Kabadiwalas waste 2–3 hours daily on inefficient routes through areas with no material to collect
3) Recyclable waste gets thrown into the household bin because the pickup never came and ends up in a landfill
4) Kabadiwalas have no digital identity, no customer record, and no way to grow their business

The Core Gap

The informal recycling system works it just has no coordination layer. Nobody has built a simple tool to connect households to their local kabadiwala before the waste gets thrown away.

Scale of the Problem

Metric	Data
Informal scrap dealers in India	~1.5 million (kabadiwalas)
Annual recyclable waste lost to landfills	~15–20 million tonnes/year
Average kabadiwala daily earnings	Rs. 400–700/day (informal estimate)
Households actively separating dry waste	<10% in tier-2/3 cities

2. User Research

This solution is designed around two user groups with distinct needs:

User Group A (Households)

1) Pain point: Stockpile newspapers, bottles, and cardboard for weeks, then throw it away in frustration when no kabadiwala arrives
2) Behavior: Most households would separate recyclables if pickup was predictable. The motivation exists the trigger does not
3) Tech level: Comfortable with WhatsApp. Use it daily for reminders and group messages
4) Key insight: They don’t want an app.They want a reminder the night before pickup

User Group B (Kabadiwalas)

5) Pain point: 30 - 40% of working time spent traveling to areas with little or no scrap to collect
6) Behavior: Already use WhatsApp for coordinating with scrap aggregators and other dealers
7) Tech level: Comfortable with voice messages and WhatsApp. Not comfortable with complex English apps
8) Key insight: They need a digital customer list and a smarter route not a full logistics platform

3. The Solution : ScrapLink

ScrapLink is a WhatsApp first coordination platform connecting households to their nearest kabadiwala giving households predictable pickup and giving kabadiwalas a digital customer base and optimized route.

How It Works

Step	Who	What Happens
1	Kabadiwala	Registers on ScrapLink via WhatsApp (name, PIN code, weekly route days). Takes 3 minutes.
2	Household	Sends their locality to ScrapLink on WhatsApp. Gets matched to nearest registered kabadiwala.
3	ScrapLink System	Night before visit: household gets a WhatsApp reminder: “Ramesh Bhai will visit your area tomorrow between 9–11am.”
4	Kabadiwala	Gets a consolidated list of registered households on their route. System suggests optimal stop order to minimize travel time.
5	Both Parties	After pickup, household rates the interaction. Kabadiwala builds a verifiable work identity over time.

Key Design Principles

1) Zero app install required runs entirely on WhatsApp Business API
2) Works on 2G and basic Android phones
3) Hindi/regional language support via WhatsApp language settings
4) No reading required for kabadiwala all system prompts are voice-message compatible

Revenue Model — The CSR Data Layer

Every kg of waste diverted from landfill is tracked. ScrapLink aggregates this data by PIN code and sells monthly impact reports to:

1) Corporate CSR teams who need verifiable waste diversion data for sustainability reports
2) Municipal corporations tracking informal sector recycling contribution
3) EPR (Extended Producer Responsibility) compliance teams companies legally required to track packaging waste recovery

This makes ScrapLink self-sustaining without charging kabadiwalas or households a single rupee.

4. Execution Plan

Phase 1

The focus is on setting up a basic communication system that allows households to register and receive reminders about the kabadiwala's collection day. A small pilot is conducted with one kabadiwala and around 10–15 households in a single locality to check whether people keep their scrap ready and whether the kabadiwala saves time.

Phase 2

once more kabadiwalas join, the system improves efficiency by organizing the daily route so that collections happen in the best possible order. The kabadiwala also provides simple feedback after each visit, which helps understand how well the system is working and where improvements are needed.

Phase 3

the focus shifts to measuring impact. The kabadiwala records the approximate amount of scrap collected, and this data is used to create a simple public report showing how much waste is being collected in each locality over time. This information can then be used to approach companies or organizations for support and to expand the project to more areas.

5. Why It Works

Practicality

1) No behavior change required kabadiwalas already do routes households already stockpile scrap
2) WhatsApp is the default communication platform for both user groups zero adoption friction
3) Can be built by a first-year student with basic Python knowledge in under 2 weeks
4) Does not require government approvals, physical infrastructure, or upfront capital

Why Existing Solutions Have Failed

1) Apps like Kabadiwala.com and Scrapuncle require smartphone downloads kabadiwalas don't install apps
2) B2B scrap platforms skip the household entirely the last-mile coordination problem remains unsolved
3) Government waste apps focus on formal municipal collection, not the informal sector

6. Impact

Direct Beneficiaries

1) Kabadiwalas: 20–30% reduction in wasted travel time, higher earnings per day, and a digital work identity
2) Households: Predictable pickup means recyclables no longer go to the trash bin
3) Environment: Measurable, verifiable increase in dry waste diverted from landfills

Scale Potential

A single WhatsApp number can serve an entire city. Onboarding one kabadiwala activates 50–100 households on their existing route. At 100 kabadiwalas onboarded, ScrapLink serves 5,000–10,000 households with no additional infrastructure.

Quantifiable Impact Metrics

1) Kg of recyclable waste diverted per month per locality
2) Average kabadiwala travel time saved per week
3) Number of households with predictable pickup before vs. after
4) CO2 equivalent avoided

7. Demo prototype